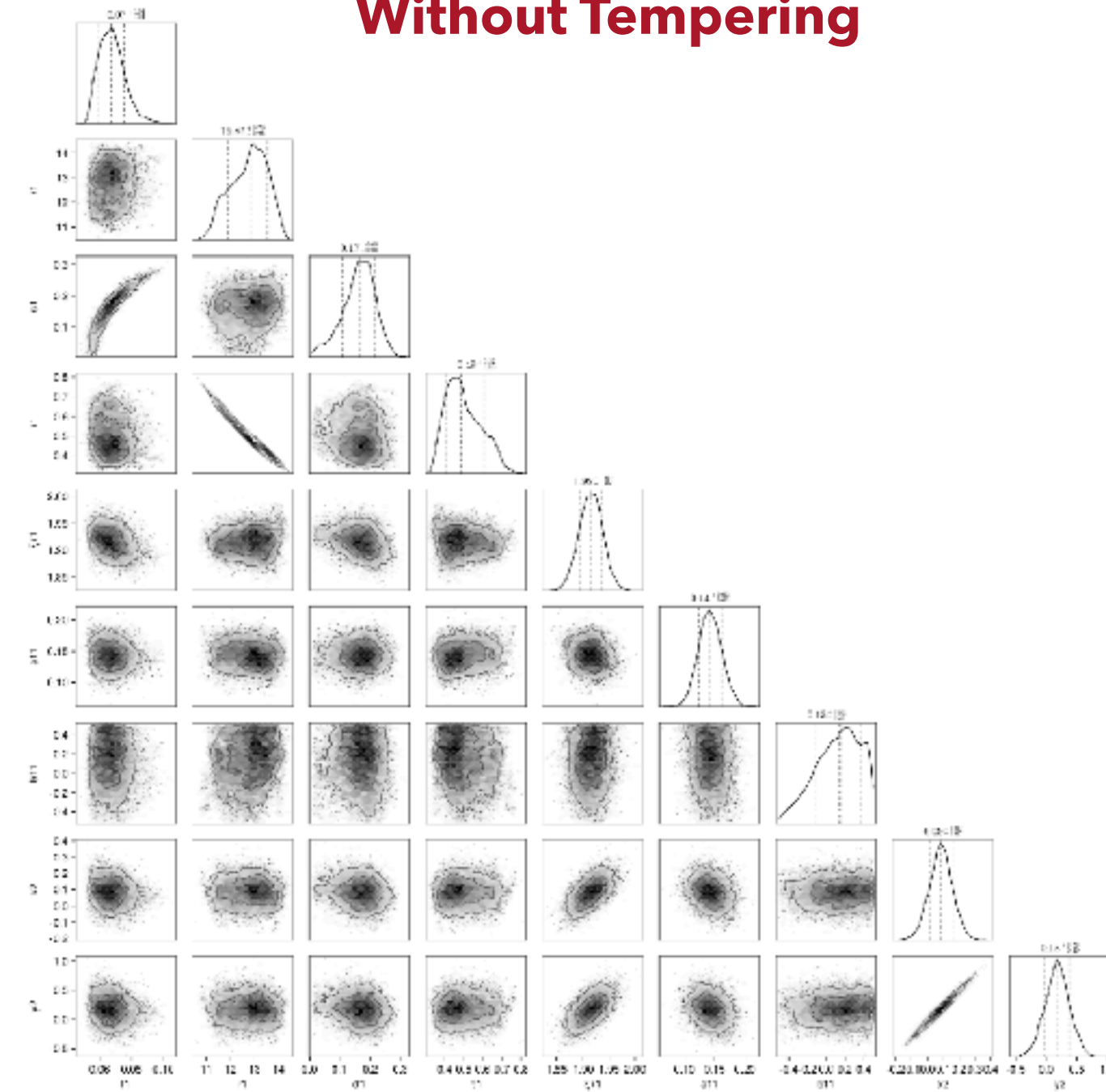
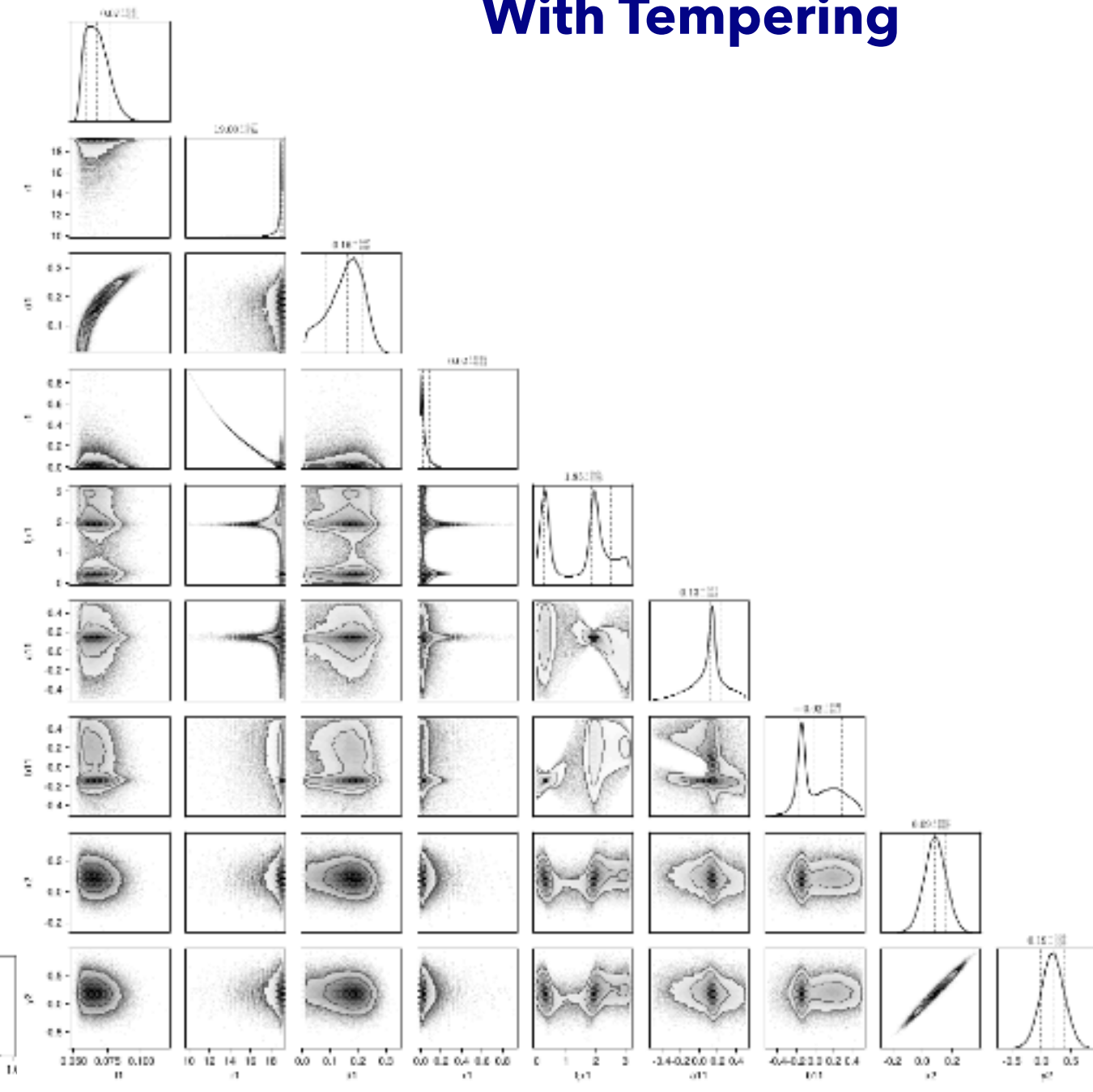


EXAMPLE: MULTIMODAL TARGETS IN ASTRONOMY

Without Tempering



With Tempering



RECENT APPLICATIONS OF NRPT

- **Cancer:** Model evolutionary dynamics of single-cell cancer genomes.
BC Cancer Research Institute & Sloan Kettering Cancer Center (Dorri et al. 2020, Salehi et al., 2021 in Nature)
- **Epidemiology:** Estimation of allele frequencies/study genetic variations in malaria.
(Murphy & Greenhouse., 2024)
- **Nuclear Fusion:** Model plasma dynamics inside worlds largest FRC nuclear fusion reactor.
Google Research & TAE Technologies (Gota et al., 2022, Langmore et al. 2023)
- **Political Science:** Identify gerrymandering in Georgia's 2021 congressional district plan
Georgia Tech (Zhao et al., 2022, Chuang et al 2024)
- **Physics:** Detect and classify orbit of brown dwarfs
(Jerry W. Xuan, 2024 in Nature)
- **Linguistics:** study the meanings of ancient Greek words change with time *(Zafar & Nicholls, 2024)*
- **Probabilistic programming:** Inference engine for various PPL/Packages in different languages:
Pigeons.jl, Blang, Comrade.jl, Themis, DrJacoby, MOIRE, Turing.jl, tensorflow probability, Hopsy

